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Furosemide: Drug information

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For additional information see "[Furosemide: Patient drug information](#)" and "[Furosemide: Pediatric drug information](#)"

For abbreviations, symbols, and age group definitions [show table](#)

ALERT: US Boxed Warning

Fluid/electrolyte loss (oral solution, tablet):



Furosemide is a potent diuretic that, if given in excessive amounts, can lead to a profound diuresis with water and electrolyte depletion. Therefore, careful medical supervision is required and dose and dose schedule must be adjusted to the individual patient's needs.

Brand Names: US

Furoscix; Lasix; Lasix ONYU

Brand Names: Canada

APO-Furosemide; BIO-Furosemide [DSC]; Furosemide Special; Lasix Oral; Lasix Special; MINT-Furosemide; PRO-Furosemide; TEVA-Furosemide

Pharmacologic Category

Antihypertensive; Diuretic, Loop

Dosing: Adult

Dosage guidance:

Dosing: Loop diuretic approximate oral dose equivalency for patients with normal kidney function: Furosemide 40 mg = bumetanide 1 mg = torsemide 10 to 20 mg ([Ref](#)).

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Ascites due to cirrhosis



Ascites due to cirrhosis:

Note: Generally used in combination with spironolactone but may be used as monotherapy for patients with hyperkalemia. For combination therapy, a dosing ratio of spironolactone 100 mg to furosemide 40 mg should generally be maintained but can be adjusted for electrolyte abnormalities ([Ref](#)).

Oral: Initial: 40 mg once daily; titrate every 3 to 5 days based on response and tolerability; once-daily dosing is preferred; maximum dose: 160 mg once daily ([Ref](#)). For small-volume ascites in patients who weigh <50 kg, some experts recommend a starting dose of 20 mg once daily ([Ref](#)).

Edema or volume overload



Edema or volume overload:

Oral, IV:

Naive to loop diuretics:

Oral, IV: Initial: 20 to 40 mg once then titrate as needed to an effective dose (see "Titration to effect" below) ([Ref](#)).

Note: Oral bioavailability varies widely but on average is 50% of the IV dose ([Ref](#)).

Refractory edema or acute decompensation in patients taking oral loop diuretics:

Bolus/intermittent dosing: IV: Initial: Administer 1 to 2.5 times the total daily oral maintenance dose once (eg, a patient taking oral furosemide 40 mg twice daily at home [80 mg/day] can be given 80 mg to 200 mg IV as an initial bolus) then titrate as needed to an effective dose; maximum effective single dose: 80 to 200 mg depending on kidney function (see "Titration to effect" below) ([Ref](#)).

Titration to effect: If the initial dose does not result in diuresis, double the individual dose (rather than administer the same dose more frequently) until diuresis occurs. Titration of an IV dose can occur at ≥ 2 -hour intervals as needed in hospitalized patients. Once an effective dose is identified, it is typically administered once or twice daily but may be given more frequently if needed. The maximum effective dose varies by population; higher-than-usual

doses may be required for patients with nephrotic syndrome or kidney failure; maximum effective single dose: 80 to 200 mg; maximum recommended total daily dose: 600 mg/day ([Ref](#)).

Continuous infusion: IV: Note: Reserve for patients who have responded to bolus therapy but need ongoing and sustained diuresis; administer continuous infusion immediately after effective bolus dose.

IV: Initial: 5 mg/hour; if diuretic response is not adequate, repeat IV bolus dose and increase continuous infusion to 10 mg/hour; continue to bolus and titrate infusion as needed up to 40 mg/hour ([Ref](#)). **Note:** In patients with eGFR <30 mL/minute/1.73m², higher doses may be necessary (see “Dosing: Altered Kidney Function: Adult”).

Note: Higher continuous infusion rates have been described but are not recommended due to potential for side effects; consider alternative strategies for fluid removal ([Ref](#)).

SUBQ (Furoscix, Lasix Onyu): Preprogrammed infusor delivers 30 mg over the first hour, followed by 12.5 mg per hour for the next 4 hours; total dose: 80 mg over 5 hours. **Note:** Not for chronic use; switch to oral diuretics as soon as clinically practical.

Transitioning from IV or SUBQ to oral: There is substantial variability in oral bioavailability; typically administer 1 to 2 times the IV or SUBQ dose orally (eg, total daily IV or SUBQ dose of 80 mg/day should be converted to an oral dose of 80 to 160 mg/day in 1 to 2 divided doses); monitor urine output and adjust oral dose as needed ([Ref](#)).

Dosage adjustment for concomitant therapy: Significant drug interactions exist, requiring dose/frequency adjustment or avoidance. Consult drug interactions database for more information.

Dosing: Kidney Impairment: Adult

The renal dosing recommendations are based upon the best available evidence and clinical expertise. Senior Editorial Team: Bruce Mueller, PharmD, FCCP, FASN, FNKF; Jason Roberts, PhD, BPharm (Hons), B App Sc, FSHP, FISAC; Michael Heung, MD, MS.

Altered kidney function:

IV, Oral:

eGFR ≥ 30 mL/minute/1.73 m²: No dosage adjustment necessary.

eGFR < 30 mL/minute/1.73 m²:

IV, Oral: Higher doses may be required to achieve desired diuretic response due to decreased secretion into the tubular fluid. However, single doses > 160 to 200 mg IV (or oral equivalent) are unlikely to result in additional diuretic effect ([Ref](#)).

Continuous infusion: IV: Initial: 20 mg/hour; if diuretic response is not adequate, repeat IV bolus dose and increase continuous infusion to 40 mg/hour ([Ref](#)).

SUBQ: There are no dosage adjustments provided in the manufacturer's labeling. Additional diuretic therapy may be required in patients with significantly reduced renal function; the on-body infusor is only able to deliver up to an 80 mg dose. Contraindicated in patients with anuria.

Hemodialysis, intermittent (thrice weekly): Not dialyzable ([Ref](#)).

IV, Oral:

Anuric patients: There is no expected clinical benefit; use not recommended ([Ref](#)).

Patients with residual kidney function: Dose as per eGFR < 30 mL/minute/1.73 m².

SUBQ: There are no dosage adjustments provided in the manufacturer's labeling. Additional fluid removal therapies may be required; the on-body infusor is only able to deliver up to an 80 mg dose. Contraindicated in patients with anuria.

Peritoneal dialysis: Not dialyzable ([Ref](#)).

IV, Oral:

Anuric patients: There is no expected clinical benefit; use not recommended ([Ref](#)).

Patients with residual kidney function: Dose as per eGFR < 30 mL/minute/1.73 m². **Note:** If necessary, limited (single dose) data suggest that an oral dose of 500 mg may be more effective than 250 mg in increasing urine output ([Ref](#)).

SUBQ: There are no dosage adjustments provided in the manufacturer's labeling. Additional fluid removal therapies may be required; the on-body infusor is only able to deliver up to an 80 mg dose. Contraindicated in patients with anuria.

CRRT:

Note: Drug clearance is dependent on the effluent flow rate, filter type, and method of renal replacement. Recommendations are based on high-flux dialyzers and effluent

flow rates of 20 to 25 mL/kg/hour (or ~1,500 to 3,000 mL/hour) unless otherwise noted.

IV, Oral: In general, use not recommended; fluid management can be more effectively managed using CRRT ultrafiltration ([Ref](#)).

PIRRT (eg, sustained, low-efficiency diafiltration):

Note: Drug clearance is dependent on the effluent flow rate, filter type, and method of renal replacement. Close monitoring of response and adverse reactions (eg, ototoxicity) due to drug accumulation is important.

IV, Oral:

Anuric patients: There is no expected clinical benefit; use not recommended ([Ref](#)).

Patients with residual kidney function: Dose as per eGFR <30 mL/minute/1.73 m² ([Ref](#)).

Dosing: Liver Impairment: Adult

The liver dosing recommendations are based upon the best available evidence and clinical expertise. Senior Editorial Team: Matt Harris, PharmD, MHS, BCPS, FAST, FCCP; Jeong Park, PharmD, MS, BCTXP, FCCP, FAST; Arun Jesudian, MD; Sasan Sakiani, MD.

Note: The diuretic response to furosemide decreases with severity of liver insufficiency and may be secondary to multiple mechanisms, including a decrease in kidney perfusion (eg, splanchnic pooling in portal hypertension) and reduced furosemide plasma protein binding, resulting in diminishing delivery of furosemide to the site of action in the kidney ([Ref](#)). Furosemide is minimally metabolized in the liver; therefore, furosemide accumulation is not expected with diminished liver function ([Ref](#)).

Liver impairment prior to treatment initiation:

Initial or dose titration in patients with preexisting liver cirrhosis:

Child-Turcotte-Pugh class A to C:

IV, Oral: No dosage adjustment necessary ([Ref](#)). Titrate gradually to desired response; higher doses may be required as liver disease progresses or in the setting of concomitant kidney impairment ([Ref](#)).

SUBQ:

Furoscix: Avoid use; the on-body infusor must be applied to the abdomen and has not been evaluated in patients with cirrhosis and/or ascites ([Ref](#)).

Lasix Onyu: Contraindicated in patients with hepatic cirrhosis.

Dosing: Older Adult

Oral, IV: Initial: 20 mg/day; increase slowly to desired response.

SUBQ: Refer to adult dosing.

Dosing: Pediatric

(For additional information see ["Furosemide: Pediatric drug information"](#))

Dosage guidance:

Dosing: Oral and parenteral (IV, IM) doses may not be interchangeable; due to differences in bioavailability, oral doses are typically higher than IV. Oral solution is available in multiple concentrations (8 mg/mL and 10 mg/mL); extra precautions should be taken to verify and avoid confusion between the different concentrations; dose should be clearly presented as mg (not mL). Dose equivalency for adult patients with normal renal function (approximate): Furosemide 40 mg = bumetanide 1 mg = torsemide 10 to 20 mg. **Note:** Ethacrynic acid has a relative potency of 0.7 ([Ref](#)).

Edema



Edema (diuresis):

Oral: Infants, Children, and Adolescents:

Intermittent dosing (acute): Oral: Initial: 2 mg/kg as a single dose; if ineffective, may increase in 6 to 8 hours in increments of 1 to 2 mg/kg/dose; maximum dose: 6 mg/kg/dose.

Maintenance dosing (chronic): Limited data available: Oral: Initial: 0.5 to 2 mg/kg/dose every 6 to 24 hours; usual initial adult dose: 20 to 40 mg/dose; if initial dose ineffective, may increase dose in increments of 1 to 2 mg/kg/dose; maximum daily dose: 6 mg/kg/**day**, not to exceed maximum adult daily dose: 600 mg/**day**; adjust dose to minimal effective dose for maintenance ([Ref](#)). **Note:** Smaller doses on a mg/kg basis may be needed in larger children, especially in those who are diuretic naive.

Intermittent IV, IM: Infants, Children, and Adolescents: Limited data available: Initial: 0.5 to 2 mg/kg/dose every 6 to 12 hours; usual initial adult dose: 20 to 40 mg/dose; if initial dose ineffective after 2 hours, may increase dose by 1 mg/kg/dose; maximum dose: 6 mg/kg/dose, not to exceed maximum adult dose: 200 mg/dose; adjust to minimal effective dose for maintenance ([Ref](#)). **Note:** Smaller doses on a mg/kg basis may be needed in larger children,

especially in those who are diuretic naive. Dosing in adolescents based on experience in adult and pediatric patients.

Continuous IV infusion:

Infants and Children: Limited data available: Initial: IV bolus dose of 1 to 2 mg/kg (reported range, 0.1 to 2 mg/kg) followed by continuous IV infusion of 0.05 to 0.4 mg/kg/**hour**; titrate dosage to clinical effect; doses up to 1 mg/kg/hour have been described ([Ref](#)).

Adolescents: Very limited data available; dosing in adolescents based on reported experience in adult and pediatric patients ([Ref](#)): IV bolus dose of 0.1 to 2 mg/kg; usual adult bolus dose: 40 mg over 1 to 2 minutes, followed by continuous IV infusion of 0.1 to 0.4 mg/kg/**hour**; usual adult dosing range: 10 to 40 mg/**hour**.

Dosage adjustment for concomitant therapy: Significant drug interactions exist, requiring dose/frequency adjustment or avoidance. Consult drug interactions database for more information.

Dosing: Kidney Impairment: Pediatric

IV, Oral:

There are no pediatric-specific recommendations; in adults with acute renal failure, very high doses may be necessary to initiate diuretic effect ([Ref](#)); avoid use in oliguric states.

Dialysis: Not removed by hemo- or peritoneal dialysis; supplemental dose is not necessary.

Dosing: Liver Impairment: Pediatric

IV, Oral: Diminished natriuretic effect with increased sensitivity to hypokalemia and volume depletion in cirrhosis; monitor effects, particularly with high doses.

Adverse Reactions (Significant): Considerations

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Acute kidney injury



Loop diuretics, including furosemide, may lead to **acute kidney injury** due to fluid loss ([Ref](#)).

Mechanism: Dose-related; related to the pharmacologic action (ie, volume depletion) ([Ref](#)).

Risk factors:

- Excessive doses ([Ref](#))
- Concurrent administration of nephrotoxic agents ([Ref](#))
- Older adults ([Ref](#))
- Preexisting volume depletion ([Ref](#))
- Reduced blood flow to the kidney or depletion of effective blood volume (eg, bilateral renal artery stenosis, cirrhosis, nephrotic syndrome, heart failure) ([Ref](#))
- Critically ill ([Ref](#))

Fluid/electrolyte loss

Loop diuretics, including furosemide, may lead to profound diuresis (especially if given in excessive amounts), resulting in **hypovolemia** and electrolyte loss. Electrolyte disturbances (eg, **hypocalcemia**, **hypokalemia**, **hypomagnesemia**) may predispose a patient to serious cardiac arrhythmias.

Mechanism: Dose-related; related to the pharmacologic action ([Ref](#)).

Risk factors:

- Excessive doses with initiation or dose adjustment ([Ref](#))
- Reduced dietary fluid and/or electrolyte intake ([Ref](#))
- Concurrent illness leading to excessive fluid loss (eg, diarrhea, vomiting) ([Ref](#))
- Concomitant administration of an additional diuretic ([Ref](#))
- Very high or very restricted dietary sodium ([Ref](#))

Hypersensitivity reactions (immediate and delayed)

Immediate hypersensitivity reactions, including **angioedema**, **urticaria**, and **anaphylaxis** have been reported with furosemide ([Ref](#)). Delayed hypersensitivity reactions ([Ref](#)) range from maculopapular **skin rash** to rare severe cutaneous adverse reactions (SCARs), including **acute generalized exanthematous pustulosis** ([Ref](#)), **drug reaction with eosinophilia and systemic symptoms** ([Ref](#)), **Stevens-Johnson syndrome** ([Ref](#)) and **toxic epidermal necrolysis**.

Mechanism: Non-dose-related; immunologic. Immediate hypersensitivity reactions: IgE-mediated, with specific antibodies formed against a drug allergen following initial exposure ([Ref](#)). SCARs: Delayed type IV hypersensitivity reactions involving a T-cell mediated drug-specific immune response ([Ref](#)).

Onset: Immediate hypersensitivity reactions: Rapid; generally occurs within 1 hour of administration but may occur up to 6 hours after exposure ([Ref](#)). Delayed hypersensitivity reactions: Varied; typically occur days to 12 weeks after drug exposure ([Ref](#)) but may occur more rapidly (usually within 1 to 4 days) upon reexposure ([Ref](#)).

Risk factors:

- Cross-reactivity: Cross-reactivity between antibiotic sulfonamides and nonantibiotic sulfonamides (such as furosemide) may not occur, or at the very least this potential is extremely low ([Ref](#)). Cross-reactivity due to antibody production (anaphylaxis) is unlikely to occur with antibiotic sulfonamides and nonantibiotic sulfonamides ([Ref](#)). There is limited published information regarding cross-reactivity between furosemide and other sulfonamides ([Ref](#)) and among other sulfonamide loop diuretics ([Ref](#)).

Ototoxicity



Loop diuretics, including furosemide, have been associated with hearing loss (**deafness**) and **tinnitus**, which is generally reversible (lasting from 30 minutes to 24 hours after administration ([Ref](#))). Cases of permanent hearing loss have also been reported ([Ref](#)).

Mechanism: Dose-related; related to the pharmacologic action (ie, inhibition of a secretory isoform of the Na-K-2Cl co-transporter in the inner ear and impacts on ionic composition of cochlear fluids) ([Ref](#)).

Risk factors:

- Concurrent kidney disease ([Ref](#))
- Excessive doses
- IV administration ([Ref](#)); bolus (higher risk) versus continuous infusion ([Ref](#))
- Concurrent use of other ototoxic agents (eg, aminoglycosides) can lead to ototoxicity at lower doses ([Ref](#))

- Premature very low birth weight (VLBW) neonates due to immature kidney function (ie, PMA <31 weeks with doses given more frequently than every 24 hours; PMA ≥31 weeks with dose given more frequently than every 12 hours) ([Ref](#)).

Adverse Reactions

The following adverse drug reactions and incidences are derived from product labeling unless otherwise specified.

Frequency not defined:

Cardiovascular: Necrotizing angiitis ([Ref](#)), orthostatic hypotension, thrombophlebitis

Dermatologic: Acute generalized exanthematous pustulosis ([Ref](#)), bulla (hemorrhagic) ([Ref](#)), bullous pemphigoid ([Ref](#)), erythema multiforme ([Ref](#)), exfoliative dermatitis, lichenoid eruption ([Ref](#)), pruritus, skin photosensitivity ([Ref](#)), skin rash, Stevens-Johnson syndrome ([Ref](#)), toxic epidermal necrolysis, urticaria ([Ref](#))

Endocrine & metabolic: Glycosuria, hyperglycemia, hyperuricemia, hypocalcemia, hypochloremic alkalosis, hypokalemia, hypomagnesemia, hypovolemia, increased serum cholesterol, increased serum triglycerides

Gastrointestinal: Abdominal cramps, anorexia, constipation, diarrhea, gastric irritation, nausea, oral irritation, pancreatitis, vomiting

Genitourinary: Bladder spasm

Hematologic & oncologic: Agranulocytosis, anemia, aplastic anemia, hemolytic anemia ([Ref](#)), leukopenia ([Ref](#)), purpuric disease, thrombocytopenia ([Ref](#))

Hepatic: Hepatic encephalopathy, increased liver enzymes, intrahepatic cholestatic jaundice

Hypersensitivity: Anaphylactic shock, anaphylaxis ([Ref](#)), angioedema ([Ref](#)), nonimmune anaphylaxis

Immunologic: Drug reaction with eosinophilia and systemic symptoms ([Ref](#))

Nervous system: Dizziness, headache, paresthesia, restlessness, vertigo

Neuromuscular & skeletal: Asthenia, muscle spasm

Ophthalmic: Blurred vision, xanthopsia

Otic: Deafness ([Ref](#)), tinnitus ([Ref](#))

Renal: Acute kidney injury, calcium nephrolithiasis (pediatric patients), interstitial nephritis (allergic) ([Ref](#)), nephrolithiasis (pediatric patients)

Miscellaneous: Fever ([Ref](#))

Postmarketing:

Endocrine & metabolic: Exacerbation of diabetes mellitus, hyponatremia (in combination with spironolactone in patients with heart failure) ([Ref](#))

Neuromuscular & skeletal: Tetany

Contraindications

Hypersensitivity to furosemide or any component of the formulation; hypersensitivity to medical adhesives (Furoscix, Lasix Onyu); anuria; hepatic cirrhosis (Lasix Onyu).

Canadian labeling: Additional contraindications (not in US labeling): Hypersensitivity to sulfonamide-derived drugs; complete kidney shutdown; hepatic coma and precoma; uncorrected states of electrolyte depletion, hypovolemia, dehydration, or hypotension; jaundiced newborn infants or infants with disease(s) capable of causing hyperbilirubinemia and possibly kernicterus; breastfeeding. **Note:** Manufacturer labeling for Lasix Special and Furosemide Special Injection also includes: GFR <5 mL/minute or GFR >20 mL/minute; hepatic cirrhosis; kidney failure accompanied by hepatic coma and precoma; kidney failure due to poisoning with nephrotoxic or hepatotoxic substances; pediatric patients ≤15 years of age.

Note: Although the FDA-approved product labeling states this medication is contraindicated in patients with hypersensitivity to sulfonamide-containing drugs, the scientific basis of this cross-sensitivity has been challenged.

Warnings/Precautions

Concerns related to adverse effects:

- Hyperuricemia: Asymptomatic hyperuricemia has been reported with use; rarely, may precipitate gout.
- Sulfonamide ("sulfa") allergy: The approved product labeling for many medications containing a sulfonamide chemical group includes a broad contraindication in patients with a prior allergic reaction to sulfonamides. There is a potential for cross-reactivity between members of a specific class (eg, two antibiotic sulfonamides). However, concerns for cross-reactivity have previously extended to all compounds containing the sulfonamide structure (SO₂NH₂). An expanded understanding of allergic mechanisms indicates cross-reactivity between antibiotic sulfonamides and

nonantibiotic sulfonamides may not occur or at the very least this potential is extremely low (Brackett 2004; Johnson 2005; Slatore 2004; Tornero 2004). In particular, mechanisms of cross-reaction due to antibody production (anaphylaxis) are unlikely to occur with nonantibiotic sulfonamides. T-cell-mediated (type IV) reactions (eg, maculopapular rash) are not well understood and it is not possible to completely exclude this potential based on current insights. In cases where prior reactions were severe (Stevens-Johnson syndrome/toxic epidermal necrolysis), some clinicians choose to avoid exposure to these classes.

- Thyroid effects: Doses >80 mg may result in transient increase in free thyroid hormones, followed by an overall decrease in total thyroid hormone levels.

Disease-related concerns:

- Adrenal insufficiency: Avoid use of diuretics for treatment of elevated blood pressure in patients with primary adrenal insufficiency (Addison disease). Adjustment of glucocorticoid/mineralocorticoid therapy and/or use of other antihypertensive agents is preferred to treat hypertension (Bornstein 2016; Inder 2015).
- Bariatric surgery: Dehydration: Avoid diuretics in the immediate postoperative period after bariatric surgery; electrolyte disturbances and dehydration may occur. Diuretics may be resumed, if indicated, once oral fluid intake goals are met (Ziegler 2009).
- Cirrhosis: In cirrhosis, avoid electrolyte and acid/base imbalances that might lead to hepatic encephalopathy; correct electrolyte and acid/base imbalances prior to initiation when hepatic coma is present. Supplemental potassium or an aldosterone antagonist, when appropriate, may reduce risk of hypokalemia and metabolic alkalosis. Close monitoring warranted, especially with initiation of therapy.
- Diabetes: Use with caution in patients with prediabetes or diabetes mellitus; may see a change in glucose control.
- Hepatic impairment: Use with caution in cirrhosis and ascites due to increased risk of hepatic encephalopathy and coma with sudden alterations of fluid and electrolytes; initiate with conservative doses and close monitoring.
- Prostatic hyperplasia/urinary stricture: May cause urinary retention due to increased urine production, especially upon initiation of therapy.
- Systemic lupus erythematosus: May cause systemic lupus erythematosus exacerbation or activation.

Special populations:

- Pediatric: May lead to nephrocalcinosis or nephrolithiasis in premature infants and in infants and children <4 years of age with chronic use. May prevent closure of patent ductus arteriosus in premature infants.
- Surgical patients: If given the morning of surgery, furosemide may render the patient volume depleted and blood pressure may be labile during general anesthesia.

Dosage form specific issues:

- On-body infusor (Furoscix, Lasix Onyu): Use in a setting where the patient can limit their activity for the duration of administration; certain movements may cause interruption of device adherence to skin and premature discontinuation of the infusion; use only in settings where alarms can be detected and responded to in order to ensure completion of dose. Not for chronic use; switch to oral diuretics as soon as clinically practical. Do not use in an MRI setting. Do not allow to get wet from water or any other fluids (eg, blood, drug products); fluid contact may lead to device errors and premature discontinuation of the infusion.
- Propylene glycol: Some dosage forms may contain propylene glycol; large amounts are potentially toxic and have been associated hyperosmolality, lactic acidosis, seizures, and respiratory depression; use caution (AAP 1997; Zar 2007).

Other warnings and precautions:

- Diuretic resistance: For some patients, despite high doses of loop diuretic, an adequate diuretic response cannot be attained. Diuretic resistance may be overcome by IV rather than oral administration or the use of two diuretics together (eg, a loop diuretic in combination with a thiazide diuretic). When multiple diuretics are used, serum electrolytes need to be monitored even more closely (ACC [Hollenberg 2019]).

Warnings: Additional Pediatric Considerations

Furosemide stimulates prostaglandin E₂ (PGE₂) which may prevent closure of patent ductus arteriosus (PDA) in premature infants. However, one large, retrospective cohort study involving 43,576 VLBW infants (median birth weight and GA: 1,120 g and 29 weeks) evaluated the association between the exposure to furosemide and the occurrence of PDA. Exposure to furosemide was not associated with an increased odds of PDA treatment (Thompson 2018). Another smaller placebo-controlled study (n=68, GA: <34 weeks, birth weight: <2,000 g) found no difference in PDA closure rate between patients treated with furosemide or placebo in combination with indomethacin (Lee 2010).

Dosage Forms: US

Excipient information presented when available (limited, particularly for generics); consult specific product labeling.

Cartridge Kit, Subcutaneous:

Furoscix: 80 mg/10 mL (1 ea)

Cartridge Kit, Subcutaneous [preservative free]:

Lasix ONYU: 80 mg/2.67 mL (1 ea) [latex free]

Solution, Injection:

Generic: 10 mg/mL (2 mL, 4 mL, 10 mL)

Solution, Injection [preservative free]:

Generic: 10 mg/mL (2 mL, 4 mL, 10 mL, 50 mL, 100 mL)

Solution, Oral:

Generic: 8 mg/mL (500 mL); 10 mg/mL (60 mL, 120 mL)

Tablet, Oral:

Lasix: 20 mg

Lasix: 40 mg, 80 mg [scored]

Generic: 20 mg, 40 mg, 80 mg

Generic Equivalent Available: US

May be product dependent

Pricing: US**Cartridge Kit** (Furoscix Subcutaneous)

80 mg/10 mL (per each): \$1,176.71

Cartridge Kit (Lasix ONYU Subcutaneous)

80MG/2.67ML (per each): \$570.00

Solution (Furosemide Injection)

10 mg/mL (per mL): \$0.30 - \$1.06

Solution (Furosemide Oral)

8 mg/mL (per mL): \$0.10

10 mg/mL (per mL): \$0.15

Tablets (Furosemide Oral)

20 mg (per each): \$0.04 - \$0.51

40 mg (per each): \$0.05 - \$0.59

80 mg (per each): \$0.08 - \$1.57

Tablets (Lasix Oral)

20 mg (per each): \$0.94

40 mg (per each): \$1.32

80 mg (per each): \$2.13

Disclaimer: A representative AWP (Average Wholesale Price) price or price range is provided as reference price only. A range is provided when more than one manufacturer's AWP price is available and uses the low and high price reported by the manufacturers to determine the range. The pricing data should be used for benchmarking purposes only, and as such should not be used alone to set or adjudicate any prices for reimbursement or purchasing functions or considered to be an exact price for a single product and/or manufacturer. Medi-Span expressly disclaims all warranties of any kind or nature, whether express or implied, and assumes no liability with respect to accuracy of price or price range data published in its solutions. In no event shall Medi-Span be liable for special, indirect, incidental, or consequential damages arising from use of price or price range data. Pricing data is updated monthly.

Dosage Forms: Canada

Excipient information presented when available (limited, particularly for generics); consult specific product labeling. [DSC] = Discontinued product

Solution, Injection:

Generic: 10 mg/mL (2 mL, 4 mL, 25 mL, 30 mL)

Solution, Intravenous:

Generic: 10 mg/mL ([DSC])

Solution, Oral:

Lasix Oral: 10 mg/mL (120 mL) [contains alcohol, usp, methylparaben (methyl p-hydroxybenzoate), polysorbate 80]

Tablet, Oral:

Lasix Special: 500 mg [contains fd&c yellow #6 (sunset yellow), quinoline yellow (d&c yellow #10)]

Generic: 20 mg, 40 mg, 80 mg

Administration: Adult

Oral: May administer with or without food.

Enteral feeding tube:

The following recommendations are based upon the best available evidence and clinical expertise. Senior editorial team: Joseph I. Boullata, PharmD, RPh, CNS-S, FASPEN, FACN; Peggi A. Guenter PhD, RN, FASPEN; Kathleen Gura, PharmD, BCNSP, FASHP, FASPEN, FPPA, FMSHP; Mark G. Klang MS, RPh, BCNSP, PhD, FASPEN; Linda Lord, NP, ACNP-BC, CNSC, FASPEN.

Oral solution (commercially available):

Note: Enteral feeding tube administration utilizing furosemide oral solution is not preferred due to the extremely high osmolality, and is not recommended for post-pyloric administration ([Ref](#)). When alternatives are not available and use of oral solution is deemed necessary for feeding tube administration, some institutions have successfully administered properly prepared oral solution; consider the risks vs benefits and ensure oral solution is adequately diluted prior to administration ([Ref](#)).

Gastric tubes (eg, NG, G-tube) (≥8 French): Dilute dose with at least an equivalent volume of purified water prior to administering to reduce osmolality; some experts recommend diluting in a volume of purified water that is 3 times the furosemide solution volume (eg, 10 mL furosemide solution diluted in 30 mL of purified water) ([Ref](#)). Draw up diluted solution into enteral dosing syringe and administer via feeding tube ([Ref](#)).

Dosage form information: Osmolality varies based on concentration and manufacturer; reported osmolality for some undiluted formulations ranges from ~2,100 to ~9,000 mOsm/kg ([Ref](#)); oral solutions with an osmolality >600 mOsm/kg may increase the probability of adverse GI effects (eg, diarrhea, cramping, abdominal distention, slowed gastric emptying), particularly if administered post-pylorically, inadequately diluted, and/or used in at-risk patients (eg, neonates and infants, patients with short bowel syndrome) ([Ref](#)).

General guidance: Hold enteral nutrition (EN) during furosemide administration ([Ref](#)). Flush feeding tube with an appropriate volume of purified water (eg,

15 mL) before administration ([Ref](#)). Following administration, rinse container used for preparation with purified water; draw up rinse and administer contents to ensure delivery of entire dose ([Ref](#)). Flush feeding tube with an appropriate volume of purified water (eg, 15 mL) and restart EN ([Ref](#)).

Oral tablet:

Gastric (eg, NG, G-tube) or post-pyloric (eg, J-tube) tubes: Crush tablet(s) into a fine powder and disperse in 10 to 20 mL purified water immediately prior to administration; draw up mixture into enteral dosing syringe and administer via feeding tube ([Ref](#)).

General guidance: Hold EN during furosemide administration ([Ref](#)). Flush feeding tube with an appropriate volume of purified water (eg, 15 mL) before administration ([Ref](#)). Following administration, rinse container used for preparation with purified water; draw up rinse and administer contents to ensure delivery of entire dose ([Ref](#)). Flush feeding tube with an appropriate volume of purified water (eg, 20 mL) and restart EN ([Ref](#)).

Note: Recommendations may not account for differences in inactive ingredients, osmolality, or other formulation properties that may vary among products from different manufacturers.

Parenteral: Undiluted direct IV injections may be administered at a rate of 20 to 40 mg per minute; for high doses (eg, ≥ 160 mg), consider a short-term infusion at a maximum rate of administration of 4 mg/minute; rapid administration increases the risk of ototoxicity due to the high concentrations achieved in a short period of time ([Ref](#)).

SUBQ:

Furoscix (on-body infusor): Administer in a setting where the patient can limit activity (limit bending; do not wear while riding in a car or flying) for the duration of administration and has access to a bathroom; not for use in an MRI setting. Prior to use, load the prefilled cartridge into the on-body infusor; do not use any other cartridges or medicines inside the on-body infusor. Select a clean, dry, hairless or nearly hairless area of the abdomen between the top of the beltline and the bottom of the ribcage (distance between top of beltline and bottom of ribcage should be at least 2½ inches); do not apply lotions, oils, or ointments to the area where infusor will be applied; wipe the area where on-body infusor will be applied with an alcohol wipe and allow to dry; rotate sites with each administration and do not apply to bruised, indurated, red, or tender areas. Remove the adhesive liner and apply on-body infusor so the cartridge window and indicator light can be seen. Firmly press the start button to begin administration; a solid green status light, beeping sound, and a white

plunger rod will fill the cartridge window when administration is complete (do not remove injection until complete). Keep on-body infusor dry; do not shower, swim, or do activities that cause sweating while wearing the on-body infusor. Do not allow to get wet from water or any other fluids (eg, blood, drug products); fluid contact may lead to device errors and premature discontinuation of the infusion. Do not use on-body infusor within 12 inches of mobile phones, tablets, computers, or wireless accessories (eg, TV remote control, Bluetooth computer keyboard, or Bluetooth mouse). For single use only; dispose of the on-body infusor in a sharps container after use.

Lasix Onyu infusor: Administer in a setting where the patient can limit activity (limit bending; do not wear while riding in a car or flying) for the duration of administration and has access to a bathroom (for at least 8 hours); not for use in an MRI setting. Select a clean, dry, hairless or nearly hairless area of the abdomen above the beltline and to the left or right of the belly button; do not apply lotions, oils, or ointments to the area where infusor will be applied; wipe the area where infusor will be applied with an alcohol wipe and allow to dry; rotate sites between the left and right with each administration and do not apply to skin that is irritated, bruised, indurated, red, or tender areas. Remove the adhesive liner and apply infusor so the start/stop button and status light are facing up and can be seen. Press and hold the start/stop button to begin administration (status light will change to flashing slow); a solid blue status light and a long beep will be heard when administration is complete (do not remove injection until complete). Keep infusor dry; do not shower, swim, or do activities that cause sweating while wearing the infusor. Do not allow to get wet from water or any other fluids (eg, blood, drug products); fluid contact may lead to device errors and premature discontinuation of the infusion. Do not use infusor within 12 inches of mobile phones, tablets, computers, or wireless accessories (eg, TV remote control, Bluetooth computer keyboard, or Bluetooth mouse). Use within 7 hours of assembling; infusion will last about 5 hours. For single use only; dispose of the infusor in a sharps container after use. Also refer to manufacturer labeling.

Note: When IV or oral administration is not possible, the sublingual route may be used. Place 1 tablet under tongue for at least 5 minutes to allow for maximal absorption. Patients should be advised not to swallow during disintegration time ([Ref](#)).

Preparation for Administration: Adult

IV infusion solution may be mixed in NS or D5W solution. May also be diluted for infusion to 1 to 2 mg/mL (maximum: 10 mg/mL).

SUBQ:

Furoscix (on-body infusor and prefilled cartridge): Clean the tip of the prefilled cartridge with an alcohol wipe. Only open the cartridge holder and load the prefilled cartridge within 30 minutes of beginning the infusion. The on-body infusor is turned on when the cartridge holder is opened. After that point, the infusion must be administered within 30 minutes or the on-body infusor will alarm and cannot restart. Insert the prefilled cartridge into the cartridge holder with the tip pointed down. Close the cartridge holder all the way until it is even with the rest of the on-body infusor. Do not press the blue start button until the on-body infusor is applied to the skin and the infusion is ready to begin.

Lasix Onyu (on-body infusor and prefilled cartridge): Charge the reusable unit until the battery light turns solid blue (flashing blue means it is still charging). Unplug charger from the reusable unit. Clean the tip of the prefilled cartridge with an alcohol wipe. Open the cartridge holder and load the prefilled cartridge with the silver tip in first; it is fully inserted when the bottom of the prefilled cartridge is at the tip of the arrow on the disposable unit. Slide together the white reusable unit and the green disposable unit; do not use any other cartridges or medicines inside the Lasix Onyu infusor. The infusor is ready when the blue status light is flashing fast and you hear one long beep. Do not press the start button until the on-body infusor is applied to the skin and the infusion is ready to begin. Start infusion within 7 hours of preparation; also refer to manufacturing labeling.

Usual Infusion Concentrations: Adult

IV infusion: 1 mg/mL **or** 2 mg/mL **or** undiluted as 10 mg/mL

Administration: Pediatric

The following feeding tube recommendations are based upon the best available evidence and clinical expertise. Senior editor panel: Joseph I. Boullata, PharmD, RPh, CNS-S, FASPEN, FACN; Peggi A. Guenter, PhD, RN, FASPEN; Kathleen Gura, PharmD, BCNSP, FASHP, FASPEN, FPPA, FMSHP; Mark G. Klang, MS, RPh, BCNSP, PhD, FASPEN; Linda Lord, NP, ACNP-BC, CNSC, FASPEN.

Note: Recommendations may not account for differences in inactive ingredients, osmolality, or other formulation properties that may vary among products from different manufacturers.

Oral: May administer with food or milk to decrease GI distress.

Oral solution (commercially available): Oral solution is available in multiple concentrations (8 mg/mL and 10 mg/mL); use extra caution during product selection and preparation and verify correct concentration is chosen. Administer with an

accurate measuring device (calibrated oral syringe or measuring cup); do not use a household teaspoon or tablespoon to measure dose (overdosage may occur).

Administration via feeding tube: Note: Enteral feeding tube administration utilizing furosemide oral solutions is not preferred due to the extremely high osmolality and is not recommended for post-pyloric administration ([Ref](#)). When alternatives are not available and use of oral solution is deemed necessary for feeding tube administration, some institutions have successfully administered properly diluted oral solution; consider the risks versus benefits and ensure oral solution is adequately diluted prior to administration ([Ref](#)).

Gastric (eg, NG, G-tube) (≥8 French): Dilute dose with at least an equivalent volume of purified water prior to administering to reduce osmolality ([Ref](#)); some experts recommend diluting in a volume of purified water that is 3 times the furosemide solution volume (eg, 2 mL furosemide solution diluted in 6 mL of purified water) ([Ref](#)). Draw up diluted solution into enteral dosing syringe and administer via feeding tube ([Ref](#)).

Dosage form information: Osmolality varies based on concentration and manufacturer; reported osmolality for some undiluted formulations ranges from ~2,100 to ~9,000 mOsm/kg ([Ref](#)); oral solutions with an osmolality >600 mOsm/kg may increase the probability of adverse GI effects (eg, diarrhea, cramping, abdominal distention, slowed gastric emptying), particularly if administered post-pylorically, inadequately diluted, and/or used in at-risk patients (eg, neonates and infants, patients with short bowel syndrome) ([Ref](#)).

General guidance: Hold enteral nutrition during furosemide administration ([Ref](#)). Flush feeding tube with the lowest volume of purified water necessary to clear the tube prior to administration based on size of patient and/or feeding tube (eg, neonates: 1 to 3 mL; infants and children: 2 to 5 mL; adolescents: 15 mL); refer to institutional policies and procedures ([Ref](#)). Following administration, rinse container used for preparation with purified water; draw up rinse and administer contents to ensure delivery of entire dose ([Ref](#)). Flush feeding tube with an appropriate volume of purified water and restart enteral nutrition ([Ref](#)).

Tablets:

Administration via feeding tube:

Gastric (eg, NG, G-tube) or post-pyloric tubes (eg, J-tube): Crush tablet(s) into a fine powder and disperse in 10 to 20 mL purified water immediately prior

to administration; draw up mixture into enteral dosing syringe and administer via feeding tube ([Ref](#)).

General guidance: Hold enteral nutrition during furosemide administration ([Ref](#)). Flush feeding tube with the lowest volume of purified water necessary to clear the tube prior to administration based on size of patient and/or feeding tube (eg, neonates: 1 to 3 mL; infants and children: 2 to 5 mL; adolescents: 15 mL); refer to institutional policies and procedures ([Ref](#)). Following administration, rinse container used for preparation with purified water; draw up rinse and administer contents to ensure delivery of entire dose ([Ref](#)). Flush feeding tube with an appropriate volume of purified water and restart enteral nutrition ([Ref](#)).

Parenteral:

IM: Administer undiluted; in general, reserve use for when enteral or IV administration is not clinically appropriate or feasible (eg, no IV access) and timely administration of furosemide is warranted (eg, acute pulmonary edema).

Intermittent IV: May be administered undiluted direct IV at a usual rate of 0.5 to 1 mg/kg/minute (not to exceed 4 mg/minute); in adults, 20 to 40 mg undiluted solution may be administered over 1 to 2 minutes.

Continuous IV infusion: Administer diluted or undiluted as a continuous IV infusion with the use of an infusion pump.

Preparation for Administration: Pediatric

Parenteral: IV: May be further diluted in NS or D5W to a concentration of 1 to 2 mg/mL for IV infusion; maximum concentration: 10 mg/mL ([Ref](#)). For neonates, Vermont Oxford Network recommends a standard concentration of 10 mg/mL for continuous infusion and concentrations of 2 mg/mL and 10 mg/mL for intermittent doses ([Ref](#)). In pediatric patients, ASHP recommends standard concentrations of 2 mg/mL and 10 mg/mL (undiluted) ([Ref](#)).

Usual Infusion Concentrations: Pediatric

IV infusion: 1 mg/mL **or** 2 mg/mL **or** undiluted as 10 mg/mL

Storage/Stability

Injection: Store intact vial/syringe at room temperature. Protect from light. Exposure to light may cause discoloration; do not use furosemide solutions if they have a yellow color. Furosemide solutions are unstable in acidic media, but very stable in basic media. Refrigeration may result in precipitation or crystallization; however, resolubilization at room temperature or warming may be performed without affecting the drug's stability.

Storage of solution diluted for infusion information is not provided in the manufacturer's labeling; refer to individual institutions policies and procedures.

SUBQ: On-body infusor and prefilled cartridge (Furoscix, Lasix Onyu): Store in carton until use at 20°C to 25°C (68°F to 77°F); excursions permitted to 15°C to 30°C (59°F to 89°F); do not refrigerate or freeze. Protect from light. Protect infusor from water.

Tablet, solution: Store at 25°C (77°F); excursions permitted to 15°C to 30°C (59°F to 89°F). Protect from light. Discard opened bottle of solution after 90 days (10 mg/mL concentration only).

Use: Labeled Indications

Edema or volume overload:

Furoscix: Treatment of edema in adult and pediatric patients weighing ≥ 43 kg with chronic heart failure or chronic kidney disease, including nephrotic syndrome.

Lasix and generics: Management of edema associated with heart failure, cirrhosis of the liver (ie, ascites), or kidney disease (including nephrotic syndrome); acute pulmonary edema.

Lasix Onyu: Treatment of edema in adult patients with chronic heart failure.

Medication Safety Issues

[Collapse All](#)

Sound-alike/look-alike issues:

Furosemide may be confused with famotidine, finasteride, fluconazole, FLUoxetine, fosinopril, loperamide, torsemide

Lasix may be confused with Lanoxin, Lidex, Lomotil, Lovenox, Luvox, Luxiq, Wakix

Older Adult: High-Risk Medication:

Beers Criteria: Diuretics are identified in the Beers Criteria as potentially inappropriate medications to be used with caution in patients 65 years and older due to the potential to cause or exacerbate syndrome of inappropriate antidiuretic hormone secretion (SIADH) or hyponatremia; monitor sodium concentration closely when initiating or adjusting the dose in older adults (Beers Criteria [AGS 2023]).

Loop diuretics are identified in the Screening Tool of Older Person's Prescriptions (STOPP) criteria as a potentially inappropriate medication in older adults (≥ 65

years). Some disease states of concern include hypertension, ankle edema, urinary incontinence, and aortic stenosis (O'Mahony 2023).

International issues:

Lasix [US, Canada, and multiple international markets] may be confused with Esidrex brand name for hydrochlorothiazide [multiple international markets]; Esidrix brand name for hydrochlorothiazide [Germany]; Losec brand name for omeprazole [multiple international markets]

Urex [Australia, Hong Kong, Turkey] may be confused with Eurax brand name for crotamiton [US, Canada, and multiple international markets]

Metabolism/Transport Effects

Substrate of OAT1/3; **Inhibits** MRP2

Drug Interactions

(For additional information: [Launch drug interactions program](#))

Note: Interacting drugs may **not be individually listed below** if they are part of a group interaction (eg, individual drugs within "CYP3A4 Inducers [Strong]" are NOT listed). For a complete list of drug interactions by individual drug name and detailed management recommendations, use the drug interactions program by clicking on the "Launch drug interactions program" link above.

Acebrophylline: May increase therapeutic effects of Furosemide. *Risk C: Monitor*

Ajmaline: Sulfonamides may increase adverse/toxic effects of Ajmaline. Specifically, the risk for cholestasis may be increased. *Risk C: Monitor*

Alfuzosin: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Aliskiren: May decrease serum concentrations of Furosemide. *Risk C: Monitor*

Allopurinol: Loop Diuretics may increase adverse/toxic effects of Allopurinol. Loop Diuretics may increase serum concentrations of Allopurinol. Specifically, Loop Diuretics may increase the concentration of Oxypurinol, an active metabolite of Allopurinol. *Risk C: Monitor*

Amifostine: Blood Pressure Lowering Agents may increase hypotensive effects of Amifostine. Management: When used at chemotherapy doses, hold blood pressure lowering medications for 24 hours before amifostine administration. If blood pressure lowering

therapy cannot be held, do not administer amifostine. Use caution with radiotherapy doses of amifostine. *Risk D: Consider Therapy Modification*

Amikacin Liposome (Oral Inhalation): Loop Diuretics may increase nephrotoxic effects of Amikacin Liposome (Oral Inhalation). Loop Diuretics may increase ototoxic effects of Amikacin Liposome (Oral Inhalation). *Risk C: Monitor*

Aminoglycosides: Loop Diuretics may increase adverse/toxic effects of Aminoglycosides. Specifically, nephrotoxicity and ototoxicity. *Risk C: Monitor*

Aminolevulinic Acid (Systemic): Photosensitizing Agents may increase photosensitizing effects of Aminolevulinic Acid (Systemic). *Risk X: Avoid*

Aminolevulinic Acid (Topical): Photosensitizing Agents may increase photosensitizing effects of Aminolevulinic Acid (Topical). *Risk C: Monitor*

Angiotensin II Receptor Blockers: Loop Diuretics may increase hypotensive effects of Angiotensin II Receptor Blockers. Loop Diuretics may increase nephrotoxic effects of Angiotensin II Receptor Blockers. *Risk C: Monitor*

Angiotensin-Converting Enzyme Inhibitors: Loop Diuretics may increase hypotensive effects of Angiotensin-Converting Enzyme Inhibitors. Loop Diuretics may increase nephrotoxic effects of Angiotensin-Converting Enzyme Inhibitors. *Risk C: Monitor*

Antidiabetic Agents: Hyperglycemia-Associated Agents may decrease therapeutic effects of Antidiabetic Agents. *Risk C: Monitor*

Antihypertensive Agents: Loop Diuretics may increase hypotensive effects of Antihypertensive Agents. *Risk C: Monitor*

Antipsychotic Agents (Second Generation [Atypical]): Blood Pressure Lowering Agents may increase hypotensive effects of Antipsychotic Agents (Second Generation [Atypical]). *Risk C: Monitor*

Arginine: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Arsenic Trioxide: Loop Diuretics may increase QTc-prolonging effects of Arsenic Trioxide. Loop Diuretics may increase hypotensive effects of Arsenic Trioxide. Management: When possible, avoid concurrent use of arsenic trioxide with drugs that can cause electrolyte abnormalities, such as the loop diuretics. *Risk D: Consider Therapy Modification*

Barbiturates: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Baxdrostat: Hyponatremia-Associated Agents may increase hyponatremic effects of Baxdrostat. *Risk C: Monitor*

Benperidol: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Beta2-Agonists: May increase hypokalemic effects of Loop Diuretics. *Risk C: Monitor*

Bilastine: Loop Diuretics may increase QTc-prolonging effects of Bilastine. *Risk C: Monitor*

Bile Acid Sequestrants: May decrease absorption of Loop Diuretics. *Risk C: Monitor*

Brigatinib: May decrease antihypertensive effects of Antihypertensive Agents. Brigatinib may increase bradycardic effects of Antihypertensive Agents. *Risk C: Monitor*

Brimonidine (Ophthalmic): May increase hypotensive effects of Antihypertensive Agents. *Risk C: Monitor*

Brimonidine (Topical): May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Bromperidol: May decrease hypotensive effects of Blood Pressure Lowering Agents. Blood Pressure Lowering Agents may increase hypotensive effects of Bromperidol. *Risk X: Avoid*

Cabozantinib: MRP2 Inhibitors may increase serum concentrations of Cabozantinib. *Risk C: Monitor*

Canagliflozin: May increase hypotensive effects of Loop Diuretics. *Risk C: Monitor*

Cardiac Glycosides: Loop Diuretics may increase adverse/toxic effects of Cardiac Glycosides. Specifically, cardiac glycoside toxicity may be enhanced by the hypokalemic and hypomagnesemic effect of loop diuretics. *Risk C: Monitor*

Cefazedone: May increase nephrotoxic effects of Loop Diuretics. *Risk C: Monitor*

Cefotiam: Loop Diuretics may increase nephrotoxic effects of Cefotiam. *Risk C: Monitor*

Cefpirome: Loop Diuretics may increase nephrotoxic effects of Cefpirome. *Risk C: Monitor*

Ceftizoxime: Loop Diuretics may increase nephrotoxic effects of Ceftizoxime. *Risk C: Monitor*

Cephaloridine: Loop Diuretics may increase nephrotoxic effects of Cephaloridine. Loop Diuretics may increase serum concentrations of Cephaloridine. *Risk X: Avoid*

Cephradine: May increase nephrotoxic effects of Loop Diuretics. *Risk C: Monitor*

Chloral Hydrate-Chloral Betaine: Furosemide may increase adverse/toxic effects of Chloral Hydrate-Chloral Betaine. Management: Consider alternatives to the coadministration of furosemide with chloral hydrate or chloral betaine when possible, as product labeling states concomitant use is not recommended. If combined, monitor for hypertension, tachycardia, and diaphoresis. *Risk D: Consider Therapy Modification*

CISplatin: Loop Diuretics may increase ototoxic effects of CISplatin. Loop Diuretics may increase nephrotoxic effects of CISplatin. *Risk C: Monitor*

Corticosteroids (Systemic): May increase hypokalemic effects of Loop Diuretics. *Risk C: Monitor*

CycloSPORINE (Systemic): May increase adverse/toxic effects of Loop Diuretics. Specifically, the risk for hyperuricemia and gout may be increased. *Risk C: Monitor*

Desmopressin: May increase hyponatremic effects of Loop Diuretics. *Risk X: Avoid*

Dexmethylphenidate: May decrease therapeutic effects of Antihypertensive Agents. *Risk C: Monitor*

Diacerein: May increase therapeutic effects of Diuretics. Specifically, the risk for dehydration or hypokalemia may be increased. *Risk C: Monitor*

Diazoxide (Immediate Release): May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Dichlorphenamide: Loop Diuretics may increase hypokalemic effects of Dichlorphenamide. *Risk C: Monitor*

Dofetilide: Loop Diuretics may increase QTc-prolonging effects of Dofetilide. Management: Monitor serum potassium and magnesium more closely when dofetilide is combined with loop diuretics. Electrolyte replacements will likely be required to maintain potassium and magnesium serum concentrations. *Risk D: Consider Therapy Modification*

DULoxetine: Blood Pressure Lowering Agents may increase hypotensive effects of DULoxetine. *Risk C: Monitor*

Empagliflozin: May increase hypotensive effects of Loop Diuretics. *Risk C: Monitor*

EPINEPHrine (Systemic): Diuretics may increase arrhythmogenic effects of EPINEPHrine (Systemic). Diuretics may decrease vasopressor effects of EPINEPHrine (Systemic). *Risk C: Monitor*

Ethacrynic Acid: Furosemide may increase ototoxic effects of Ethacrynic Acid. *Risk X: Avoid*

Fexinidazole: May increase serum concentrations of OAT1/3 Substrates (Clinically Relevant). Management: Avoid use of fexinidazole with OAT1/3 substrates when possible. If combined, monitor for increased OAT1/3 substrate toxicities. *Risk D: Consider Therapy Modification*

Flunarizine: May increase therapeutic effects of Antihypertensive Agents. *Risk C: Monitor*

Foscarnet: Loop Diuretics may increase serum concentrations of Foscarnet. Management: When diuretics are indicated during foscarnet treatment, thiazides are recommended

over loop diuretics. If patients receive loop diuretics during foscarnet treatment, monitor closely for evidence of foscarnet toxicity. *Risk D: Consider Therapy Modification*

Fosphenytoin-Phenytoin: May decrease diuretic effects of Furosemide. *Risk C: Monitor*

Herbal Products with Blood Pressure Increasing Effects: May decrease antihypertensive effects of Antihypertensive Agents. *Risk C: Monitor*

Herbal Products with Blood Pressure Lowering Effects: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Hypotension-Associated Agents: Blood Pressure Lowering Agents may increase hypotensive effects of Hypotension-Associated Agents. *Risk C: Monitor*

Iloperidone: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Immune Globulin: Loop Diuretics may increase thrombogenic effects of Immune Globulin. *Risk C: Monitor*

Indoramin: May increase hypotensive effects of Antihypertensive Agents. *Risk C: Monitor*

Iodinated Contrast Agents: Loop Diuretics may increase nephrotoxic effects of Iodinated Contrast Agents. *Risk C: Monitor*

Ipragliflozin: May increase adverse/toxic effects of Loop Diuretics. Specifically, the risk for intravascular volume depletion may be increased. *Risk C: Monitor*

Isocarboxazid: May increase hypotensive effects of Diuretics. *Risk X: Avoid*

Ivabradine: Loop Diuretics may increase arrhythmogenic effects of Ivabradine. *Risk C: Monitor*

Leflunomide: May increase serum concentrations of OAT1/3 Substrates (Clinically Relevant). *Risk C: Monitor*

Levodopa-Foslevodopa: Blood Pressure Lowering Agents may increase hypotensive effects of Levodopa-Foslevodopa. *Risk C: Monitor*

Levosulpiride: Loop Diuretics may increase adverse/toxic effects of Levosulpiride. *Risk X: Avoid*

Licorice: May increase hypokalemic effects of Loop Diuretics. *Risk C: Monitor*

Lithium: Loop Diuretics may decrease serum concentrations of Lithium. Loop Diuretics may increase serum concentrations of Lithium. *Risk C: Monitor*

Loop Diuretics: May increase hypotensive effects of Antihypertensive Agents. *Risk C: Monitor*

Lormetazepam: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Mecamylamine: Sulfonamides may increase adverse/toxic effects of Mecamylamine. *Risk X: Avoid*

Metergoline: May decrease antihypertensive effects of Blood Pressure Lowering Agents. Blood Pressure Lowering Agents may increase orthostatic hypotensive effects of Metergoline. *Risk C: Monitor*

Methotrexate: May decrease therapeutic effects of Loop Diuretics. Loop Diuretics may increase serum concentrations of Methotrexate. Methotrexate may increase serum concentrations of Loop Diuretics. *Risk C: Monitor*

Methoxsalen (Systemic): Photosensitizing Agents may increase photosensitizing effects of Methoxsalen (Systemic). *Risk C: Monitor*

Methylphenidate: May decrease antihypertensive effects of Antihypertensive Agents. *Risk C: Monitor*

Milsaperidone: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Molsidomine: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Naftopidil: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Netilmicin (Ophthalmic): Loop Diuretics may increase nephrotoxic effects of Netilmicin (Ophthalmic). *Risk X: Avoid*

Neuromuscular-Blocking Agents: Loop Diuretics may decrease neuromuscular-blocking effects of Neuromuscular-Blocking Agents. Loop Diuretics may increase neuromuscular-blocking effects of Neuromuscular-Blocking Agents. *Risk C: Monitor*

Nicergoline: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Nicorandil: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Nitisinone: May increase serum concentrations of OAT1/3 Substrates (Clinically Relevant). *Risk C: Monitor*

Nitroprusside: Blood Pressure Lowering Agents may increase hypotensive effects of Nitroprusside. *Risk C: Monitor*

Nonsteroidal Anti-Inflammatory Agents (Topical): May decrease therapeutic effects of Loop Diuretics. *Risk C: Monitor*

Nonsteroidal Anti-Inflammatory Agents: May decrease diuretic effects of Loop Diuretics. Loop Diuretics may increase nephrotoxic effects of Nonsteroidal Anti-Inflammatory Agents. Management: Monitor for evidence of kidney injury or decreased therapeutic effects of loop diuretics with concurrent use of an NSAID. Consider avoiding concurrent use in CHF or cirrhosis. Concomitant use of bumetanide with indomethacin is not recommended. *Risk D: Consider Therapy Modification*

Norepinephrine: Furosemide may decrease therapeutic effects of Norepinephrine. *Risk C: Monitor*

Obinutuzumab: May increase hypotensive effects of Blood Pressure Lowering Agents. Management: Consider temporarily withholding blood pressure lowering medications beginning 12 hours prior to obinutuzumab infusion and continuing until 1 hour after the end of the infusion. *Risk D: Consider Therapy Modification*

Opioid Agonists: May increase adverse/toxic effects of Diuretics. Opioid Agonists may decrease therapeutic effects of Diuretics. *Risk C: Monitor*

Palopegteriparatide: Loop Diuretics may decrease therapeutic effects of Palopegteriparatide. Loop Diuretics may increase therapeutic effects of Palopegteriparatide. *Risk C: Monitor*

Pentoxifylline: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Perazine: May increase hypotensive effects of Antihypertensive Agents. *Risk C: Monitor*

Periciazine: Antihypertensive Agents may increase orthostatic hypotensive effects of Periciazine. *Risk C: Monitor*

Pholcodine: Blood Pressure Lowering Agents may increase hypotensive effects of Pholcodine. *Risk C: Monitor*

Phosphodiesterase 5 Inhibitors: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Piperacillin: May increase hypokalemic effects of Diuretics. *Risk C: Monitor*

Polyethylene Glycol-Electrolyte Solution: Diuretics may increase nephrotoxic effects of Polyethylene Glycol-Electrolyte Solution. *Risk C: Monitor*

Porfimer: Photosensitizing Agents may increase photosensitizing effects of Porfimer. *Risk X: Avoid*

Prazosin: Antihypertensive Agents may increase hypotensive effects of Prazosin. *Risk C: Monitor*

Pretomanid: May increase serum concentrations of OAT1/3 Substrates (Clinically Relevant).
Risk C: Monitor

Probenecid: May decrease diuretic effects of Loop Diuretics. Probenecid may increase serum concentrations of Loop Diuretics. *Risk C: Monitor*

Promazine: Loop Diuretics may increase QTc-prolonging effects of Promazine. *Risk X: Avoid*

Prostacyclin Analogues: May increase hypotensive effects of Blood Pressure Lowering Agents.
Risk C: Monitor

Rasagiline: Blood Pressure Lowering Agents may increase hypotensive effects of Rasagiline.
Risk C: Monitor

Reboxetine: May increase hypokalemic effects of Loop Diuretics. *Risk C: Monitor*

Salicylates: May decrease therapeutic effects of Loop Diuretics. Loop Diuretics may increase serum concentrations of Salicylates. *Risk C: Monitor*

Selegiline: Blood Pressure Lowering Agents may increase hypotensive effects of Selegiline.
Risk C: Monitor

Semaglutide: Furosemide may decrease therapeutic effects of Semaglutide. Semaglutide may increase serum concentrations of Furosemide. *Risk C: Monitor*

Silodosin: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Sodium Phosphates: Diuretics may increase nephrotoxic effects of Sodium Phosphates. Specifically, the risk of acute phosphate nephropathy may be enhanced. *Risk C: Monitor*

Sucralfate: May decrease serum concentrations of Furosemide. Sucralfate may impair the absorption of furosemide. Management: Separate orally administered furosemide from sucralfate administration by at least 2 hours. *Risk D: Consider Therapy Modification*

Taurursodiol: May increase serum concentrations of OAT1/3 Substrates (Clinically Relevant).
Risk X: Avoid

Temoporfin: Photosensitizing Agents may increase photosensitizing effects of Temoporfin.
Risk X: Avoid

Terazosin: Antihypertensive Agents may increase hypotensive effects of Terazosin. *Risk C: Monitor*

Teriflunomide: May increase serum concentrations of OAT1/3 Substrates (Clinically Relevant).
Risk C: Monitor

Thyroid Products: Furosemide may decrease protein binding of Thyroid Products. This may lead to a transient increase in free thyroid hormone concentrations and to a later decrease in total thyroid hormone concentrations. *Risk C: Monitor*

Tobramycin (Oral Inhalation): Loop Diuretics may increase ototoxic effects of Tobramycin (Oral Inhalation). Loop Diuretics may increase nephrotoxic effects of Tobramycin (Oral Inhalation). *Risk X: Avoid*

Tolcapone: Blood Pressure Lowering Agents may increase hypotensive effects of Tolcapone.
Risk C: Monitor

Topiramate: Loop Diuretics may increase hypokalemic effects of Topiramate. *Risk C: Monitor*

Urapidil: Antihypertensive Agents may increase hypotensive effects of Urapidil. *Risk C: Monitor*

Vaborbactam: May increase serum concentrations of OAT1/3 Substrates (Clinically Relevant).
Risk C: Monitor

Vadadustat: May increase serum concentrations of OAT1/3 Substrates (Clinically Relevant).
Risk C: Monitor

Vancomycin: Loop Diuretics may increase nephrotoxic effects of Vancomycin. *Risk C: Monitor*

Verteporfin: Photosensitizing Agents may increase photosensitizing effects of Verteporfin.
Risk C: Monitor

Xipamide: May increase adverse/toxic effects of Loop Diuretics. Specifically, the risk of hypovolemia, electrolyte disturbances, and prerenal azotemia may be increased. *Risk C: Monitor*

Zoledronic Acid: Loop Diuretics may increase hypocalcemic effects of Zoledronic Acid. *Risk C: Monitor*

Reproductive Considerations

When diuretics are used for the treatment of heart failure in patients planning to become pregnant, adjust dose prior to conception to minimize risk of placental hypoperfusion (AHA/ACC/HFSA [Heidenreich 2022]).

Pregnancy Considerations

Furosemide crosses the placenta (Beerman 1978; Gonçalves 2020; Riva 1978).

Monitor fetal growth if used during pregnancy (ESC [Regitz-Zagrosek 2018]).

Due to pregnancy-induced physiologic changes, some pharmacokinetic properties of furosemide may be altered. Following administration of a single oral dose of furosemide to patients prior to cesarean delivery, the V_d and clearance were increased, and the C_{max} was lower compared to nonpregnant, healthy individuals (Gonçalves 2020).

Heart failure in pregnancy is associated with adverse maternal and fetal outcomes, including premature birth, infants born small-for-gestational-age, increased risk of maternal and fetal death (Bright 2021). Furosemide may be used for symptom management in pregnant patients with heart failure complicated by pulmonary congestion; closely monitor volume status and adjust dose to minimize risk of placental hypoperfusion (AHA/ACC/HFSA [Heidenreich 2022]; ESC [Regitz-Zagrosek 2018]).

Short-term postpartum use of furosemide is being evaluated to reduce the risk of persistent hypertension in patients who had hypertensive disorders of pregnancy (Lopes Perdigao 2021, Veena 2017); additional data needed (Malhamé 2023).

Breastfeeding Considerations

Furosemide is present in breast milk.

According to the manufacturer, the decision to breastfeed during therapy should consider the risk of infant exposure, the benefits of breastfeeding to the infant, and the benefits of treatment to the mother. In general, large doses of loop diuretics have the potential to decrease milk volume and suppress lactation; use should be avoided when possible (WHO 2002). When used for the treatment of heart failure, furosemide may be considered with close neonatal monitoring (AHA/ACC/HFSA [Heidenreich 2022]).

Dietary Considerations

May cause potassium loss; potassium supplement or dietary changes may be required.

Monitoring Parameters

BP; serum electrolytes; kidney function; fluid intake and output.

Mechanism of Action

Primarily inhibits reabsorption of sodium and chloride in the ascending loop of Henle and proximal and distal renal tubules, interfering with the chloride-binding cotransport system, thus causing its natriuretic effect (Rose 1991).

Pharmacokinetics (Adult Data Unless Noted)

Onset of action:**Diuresis:**

IV: 5 minutes.

Oral, sublingual (SL): 30 to 60 minutes.

Symptomatic improvement with acute pulmonary edema: Within 15 to 20 minutes; occurs prior to diuretic effect.

Peak effect:

Oral, SL: 1 to 2 hours.

IV:

Neonates and Infants (mean GA 30.7 weeks [range: 26 to 35 weeks], mean PNA 29 days [range: 10 to 57 days]): 1 to 3 hours (Ross 1978).

Adults: 0.5 hours.

SUBQ: 4 to 5 hours.

Duration:

Oral, SL: 6 to 8 hours.

IV:

Neonates and Infants (mean GA 30.7 weeks [range: 26 to 35 weeks], mean PNA 29 days [range: 10 to 57 days]): 6 hours (Ross 1978).

Adults: ~2 hours.

Protein binding: 91% to 99%; primarily to albumin.

Metabolism: Minimally hepatic.

Bioavailability: Oral tablet: 47% to 64%; Oral solution: 50%; SUBQ: 99.6% to 112%; SL administration of oral tablet: ~60%; results of a small comparative study (n=11) showed bioavailability of SL administration of tablet was ~12% higher than oral administration of tablet (Haegeli 2007).

Half-life elimination:**Preterm neonates:**

GA <28 weeks (PNA <14 days): IV: 15.3 ± 2.3 hours (Okazaki 2022).

GA 26 to 36 weeks (PNA <21 days): IV: 19.9 ± 3 hours (range: 8.7 to 46 hours) (Peterson 1980).

Adults: Oral, IV: Normal kidney function: 0.5 to 2 hours; End-stage kidney disease: 9 hours; SUBQ: 3.2 ± 0.9 hours.

Excretion: Urine (Oral: 50%, IV: 80%) within 24 hours; feces (as unchanged drug); nonrenal clearance prolonged in kidney impairment.

Patient Counseling Points

(For additional information see ["Furosemide: Patient drug information"](#))

What is this drug used for?

- It is used to get rid of extra fluid.
- It is used to treat high blood pressure.

All drugs may cause side effects. However, many people have no side effects or only have minor side effects. Call your doctor or get medical help if any of these side effects or any other side effects bother you or do not go away:

All products:

- Dizziness or headache
- Constipation, diarrhea, upset stomach, throwing up, or decreased appetite
- Stomach cramps

On-Body Infusor:

- Swelling, warmth, or redness where this drug was given
- Bruising, soreness, or short-term pain where this drug was given

WARNING/CAUTION: Even though it may be rare, some people may have very bad and sometimes deadly side effects when taking a drug. Tell your doctor or get medical help right away if you have any of the following signs or symptoms that may be related to a very bad side effect:

- Fluid and electrolyte problems like mood changes, confusion, muscle pain or weakness, fast or abnormal heartbeat, severe dizziness or passing out, increased thirst, seizures, feeling very tired or weak, decreased appetite, unable to pass urine or change in the amount of urine produced, dry mouth, dry eyes, or severe upset stomach or throwing up

- High blood sugar like confusion, feeling sleepy, unusual thirst or hunger, passing urine more often, flushing, fast breathing, or breath that smells like fruit
- Liver problems like dark urine, tiredness, decreased appetite, upset stomach or stomach pain, light-colored stools, throwing up, or yellow skin or eyes
- Kidney problems like unable to pass urine, change in how much urine is passed, blood in the urine, or a big weight gain
- A pancreas problem (pancreatitis) like very bad stomach pain, very bad back pain, or very bad upset stomach or throwing up
- Severe dizziness or passing out
- A burning, numbness, or tingling feeling that is not normal
- Blurred eyesight
- Restlessness
- Hearing problems like decreased hearing and loss of hearing, ringing in the ears, any change in your hearing, or a feeling of fullness in your ears
- Infection like fever, chills, or sore throat
- Any unexplained bruising or bleeding
- Feeling very tired or weak
- Severe skin reactions; red, swollen, blistered, or peeling skin; red or irritated eyes; sores in the mouth, throat, nose, eyes, genitals, or any areas of skin; fever; chills; body aches; shortness of breath; or swollen glands
- Signs of an allergic reaction, like rash; hives; itching; red, swollen, blistered, or peeling skin with or without fever; wheezing; tightness in the chest or throat; trouble breathing, swallowing, or talking; unusual hoarseness; or swelling of the mouth, face, lips, tongue, or throat.

Note: This is not a comprehensive list of all side effects. Talk to your doctor if you have questions.

Consumer Information Use and Disclaimer: This information should not be used to decide whether or not to take this medicine or any other medicine. Only the healthcare provider has the knowledge and training to decide which medicines are right for a specific patient. This information does not endorse any medicine as safe, effective, or approved for treating any

patient or health condition. This is only a limited summary of general information about the medicine's uses from the patient education leaflet and is not intended to be comprehensive. This limited summary does NOT include all information available about the possible uses, directions, warnings, precautions, interactions, adverse effects, or risks that may apply to this medicine. This information is not intended to provide medical advice, diagnosis or treatment and does not replace information you receive from the healthcare provider. For a more detailed summary of information about the risks and benefits of using this medicine, please speak with your healthcare provider and review the entire patient education leaflet.

Brand Names: International

International Brand Names by Country

For country code abbreviations ([show table](#))

(AE) United Arab Emirates: Apo furosemide | Diomide | Diuresal | Furomex | Fusix | Impugan | Lasix | Lasix faible | Oedemex | Salurin; (AR) Argentina: Dafil | Dafurose | Errolon | Fabofurox | Fada furosemida | Frecuental | Furagrand | Furix | Furosemida | Furosemida biol | Furosemida denver farma | Furosemida fecofar | Furosemida klonal | Furosemida rigo | Furosemida vannier | Furosemida vent 3 | Furotral | Fursemida | Fursemida fabra | Fursemida northia | Furtenk | Kolkin | Lasix | Nuriban | Retep | Tiuril; (AT) Austria: Furohexal | Furon | Furosemid-1a pharma gmbh | Furosemid-genericon pharma | Furostad | Furotyrol | Lasix; (AU) Australia: Cm frusemide | Frusax | Frusehexal | Frusemide | Frusemide an | Frusemide apotex | Frusemide PS | Frusemide rbx | Frusemide Sandoz | Frusemide-bc | Frusemix | Frusemix M | Frusid | Furosemid ratiopharm | Furosemide an | Lasix | Lasix m | Noumed Furosemide | Tw frusemide | Uremide | Urex; (BD) Bangladesh: Frunep | Frusemide | Frusin | Frusix | Fusid | Lasix | Renalix | Trofurit | Uratic | Urex; (BE) Belgium: Docfurose | Furodur | Furosemid ratiopharm | Furosemide eurogenerics | Furosemide sandoz | Furosemide teva generics belgium | Furotop | Lasix; (BF) Burkina Faso: Furo denk | Lasilix; (BG) Bulgaria: Furanthril | Furosemid; (BR) Brazil: Biosemide | Diflumid | Diuremida | Diuremil | Diurit | Fluxil | Furesin | Furolasil | Furomida | Furosan | Furosantisa | Furosecord | Furosem | Furosemida | Furosemil | Furosemin | Furosetron | Furosix | Furp furosemida | Fursemida | Lasiol | Lasix | Lqfex furosemida | Neosemid | Rosemid | Unifurix | Urasix; (CH) Switzerland: Furo-basan | Furodrix | Furosemid helvepharm | Furosemid UPSA | Furosemid-cophar | Furosemid-trendpharm | Furosemide zentiva | Fursol | Lasix | Oedemex; (CI) Côte d'Ivoire: Apo furosemide; (CL) Chile: Asax | Furosemida | Furosemida andromaco | Lasix | Laxur; (CN) China: Furosemide; (CO) Colombia: Askast | Diuril | Fluxor | Furomix | Furosemida | Furosemida MK | Hissuflux | Lasix; (CR) Costa Rica: Furosemida | Furosemida MK | Lasix | Solax; (CZ) Czech Republic: Apo furosemide | Dryptal | Durafurid | Frusemide | Furantral | Furantril | Furesis | Furomex | Furon | Furorese | Furosemid | Furosemid aurovitas | Furosemid fmk | Furosemid kabi | Furosemid medreg | Furosemid ratiopharm | Furosemid xantis | Furosemidum | Fursemid | Lasix; (DE) Germany: Diurapid | Durafurid | Furanthril | Furo 1 A Pharma | Furo abz | Furo biochemie | Furo CT | Furo Lut | Furobeta | Furogamma | Furomed wolff | Furorese | Furosal | Furosemid | Furosemid 1a pharm | Furosemid 3mm | Furosemid al | Furosemid denk | Furosemid Dura | Furosemid Heumann | Furosemid Heumann Heunet | Furosemid ratiopharm | Furosemid Sandoz | Furosemid stada | Fusid | Hydro-rapid-tablinen | Jufurix | Lasix | Oedemase | Sigasalur; (DK) Denmark: Diural; (DO) Dominican Republic: Furaced | Furolax | Furosemid | Furosemida | Furosemida lam | Furotil | Lasix; (EC) Ecuador: Asax | Furosemida | Furosemida Genfar | Furosemida MK | Furosemida nifa | Lasix | Odemex | Ragilon; (EE) Estonia: Frusemide | Frusol |

Furesis | Furon | Furosemid | Furosemid-alpharma | Furosemida vitoria | Furosemidum
 polpharma | Furosemiid ns | Lasix; (EG) Egypt: Frusemide | Furoretic | Lafurex | Lasix | Odement |
 Salex; (ES) Spain: Acuix | Furosemida | Furosemida alter | Furosemida aphar | Furosemida
 aurovitas | Furosemida Bexal | Furosemida bluefish | Furosemida Cinfa | Furosemida combix |
 Furosemida ges | Furosemida gesfur | Furosemida Kern Pharma | Furosemida Mabo | Furosemida
 Mylan | Furosemida Normon | Furosemida pensa | Furosemida ratiopharm | Furosemida Sala |
 Furosemida Sandoz | Furosemida stada | Furosemida tecnigen | Furosemida teva | Furosemida vir
 | Seguril; (ET) Ethiopia: Furo denk; (FI) Finland: Apores | Furesis | Furesis special | Furix | Furomex
 | Furomin | Furosal | Furotrid | Impugan | Lasix | Lasix special | Vesix | Vesix special; (FR) France:
 Furosemide arrow | Furosemide Biogaran | Furosemide dakota | Furosemide eg | Furosemide gnr
 | Furosemide irex | Furosemide merck | Furosemide ratiopharm | Furosemide rpg | Furosemide
 sandoz | Furosemide teva | Furosemide Zydus | Furosemix | Lasilix | Lasilix Faible | Lasilix special;
 (GB) United Kingdom: Aluzine | Diuresal | Dryptal | Froop | Frumax | Frusemide | Frusemide
 almus | Frusemide arrow | Frusemide berk | Frusemide cox | Frusemide dc | Frusemide kent |
 Frusemide Sandoz | Frusetic | Frusid | Frusol | Furosemide Focus | Furosemide teva | Hydroled |
 Lasix | Rusyde; (GR) Greece: Frusemide | Fudesix | Furanthrill | Furosemid | Furosemid germed |
 Furosemide teva | Hydroflux | Lasix | Salurex | Semid; (GT) Guatemala: Furo denk | Furosemida |
 Furosemida asmoh | Furosemida aurochem; (HK) Hong Kong: Apo furosemide | Diurex |
 Frusemide | Frusid | Frutin | Furosemide actavis | Fusid | Lasix | Syn-Furosemide | Urex; (HR)
 Croatia: Edemid | Fursemid | Lasix; (HU) Hungary: Furon | Furosemid | Furosemid al | Furosemid
 chinoin | Furosemid ratiopharm | Huma-semide | Lasix; (ID) Indonesia: Arsiret | Cetasix | Classic |
 Diurefo | Farsiretic | Farsix | Furosix | Gralixa | Husamid | Impugan | Lasix | Laveric | Naclex |
 Salurix | Uresix | Yekasix; (IE) Ireland: Fruside | Frusol | Lasix; (IL) Israel: Fusid | Lasix | Miphar; (IN)
 India: Alfride | Frasix | Frusemide | Frusenex | Frusix | Furoped | Lasix | Salinex | Urix; (IQ) Iraq:
 Awafurosem | Furosemide mdi | Lasimex; (IT) Italy: Furosemide Angenerico | Furosemide
 aurobindo | Furosemide doc generici | Furosemide l.f.m. | Furosemide merck generics |
 Furosemide mylan generics | Lasix; (JO) Jordan: Diusemide | Lasix; (JP) Japan: A basedock | Accent |
 Anfuramide kobayashi kako | Anfuramide merck hoei | Beronald | Depix | Diusemide | Foliront |
 Frusolon | Fulsix | Furfan | Furosedon | Furosemide | Furosemide amel | Furosemide asahikasei |
 Furosemide isei | Furosemide kayaku | Furosemide merck hoei | Furosemide mita | Furosemide
 s.s. seiyaku | Furosemide shiono | Furosemide showa | Furosemide sn | Furosemide taiyo |
 Furosemide takeda teva | Furosemide tobishi | Furosemide towa | Furosemin | Fursein |
 Fursemait | Fursemide | Furusemadone | Furusemide | Katlex | Kutrix | Lasix | Lowpston | Luseck
 | Macasirool | Maoread | Moilarorin | Neudiurese | Polysquall | Polysquall a | Profemin |
 Promedes | Protargen | Radiamin hexal | Radiamin nippon | Radonna | Rasisemide | Rosemide |
 Synephron | Tabilon a | Urenil | Urex | Urosemide; (KE) Kenya: Asix | Carefusem | Diuresal |
 Frusemide | Furo denk | Furomed | Fusid | Lasix | Lefrusid; (KR) Korea, Republic of: Dailix | Dairix |
 Furix | Lasix | Laxed | Rasidine; (KW) Kuwait: Furosemide sandoz | Fusix | Lasix; (LB) Lebanon: Apo
 Furosemid | Diuresal | Furesix | Furosemid Genericon | Furosetic | Lasix; (LT) Lithuania: Apo
 Furosemid | Diurapid | Dryptal | Durafurid | Furanthrill | Furesis | Furo puren | Furomex | Furon |
 Furoped | Furosemid | Furosemid bp | Fursemid | Fusid | Lasix; (LU) Luxembourg: Docfurose |
 Furodur | Furorese | Lasix; (LV) Latvia: Apo Furosemid | Dryptal | Durafurid | Frusemide | Frusix |
 Furanthrill | Furesis | Furo CT | Furomex | Furon | Furosemid | Fursemid | Lasix; (MA) Morocco:
 Duresan | Furilan | Lasilix | Lasilix special; (MX) Mexico: Biomisen | Diufamin | Diurazol |
 Diurmessel | Diusix | Dromirix | Frudemisan | Furoliz | Furosan | Furosemida | Furosemida
 hormona | Furosemida iqfa | Furosemida kendrick | Furoter | Klimyn | Lasix | Ordafil | Osemin |
 Semik; (MY) Malaysia: Dirine | Diuremid | Frucid | Frumid | Frusemide | Frusemide Pharmaniaga |
 Fruside | Frusix | Furmide | Fusix | Lasix | Rasitol | Trofurit | Usix | Vusimide; (NG) Nigeria: Bakan
 gizo furosemide | Baraxix | Frucare | Frusatex | Jeklargzis | Lafest | Lasimac | Lasinex | Renix |
 Shree axis | Tamar & pharez furosemide | Wellcare furosemide; (NL) Netherlands: Furosemide |

Furosemide Alparma | Furosemide cf | Furosemide flx | Furosemide ratiopharm | Furosemide sandoz | Furosemide xiromed | Furosemidum | Lasiletten | Lasix; (NO) Norway: Diural | Frusol | Furix | Furosemid | Lasix | Lasix liquidum; (NZ) New Zealand: Apo furosemide | Apo-frusemide | Diurin | Frusemide | Frusid | Lasix | Urex Forte; (PA) Panama: Duriten | Furophar | Furosem | Furosemida | Furosemida Cinfa | Furosemida Kern Pharma | Furosemida MK | Lasix; (PE) Peru: Diufamin | Frusamil | Furosemida | Fusemi | Lasix | Ragilon; (PH) Philippines: Dimide | Diurem | Diurema | Diuril | Diospec | Edemann | Fretic | Frusema | Furix | Furosemide | Fusem | Indirex | Kenzamide | Lasix | Marsemide | Pharmix | Pisamor | Ritemed frusema | Tgp diuretab | Uromid | Veramide; (PK) Pakistan: Aquasin | Asumide | Basix | Diurex | Frusemide | Frusol | Genmide | Lasix | Laxide | Losamide | Odenil | Uremide; (PL) Poland: Desal | Furantril | Furosemide medreg | Furosemidum | Furosemidum polpharma | Lasilix | Lasix; (PR) Puerto Rico: Furoscix | Furosemide | Lasix; (PT) Portugal: Entix | Furosemida | Furosemida aurovitas | Furosemida bluefish | Furosemida farmoz | Furosemida Generis | Furosemida labesfal | Furosemida Pharmakern | Furosemida Sandoz | Furosemida tolife | Furosemida Zentiva | Lasix | Naqua; (PY) Paraguay: Diuride | Fabofurox | Furosemida | Furosemida heisecke | Hidro max | Lasix | Ragilon | Solax | Urimeg; (QA) Qatar: Desal | Diusemide | Fuomid | Furosemid | Furoxem | Fusix | Lasemid | Lasix | Lasix High Dose | Salurex | Salurin; (RO) Romania: Furantril | Furomex | Furosemid arena | Furosemid laropharm | Furosemid lph | Furosemid medreg | Furosemid slavia | Furosemid Zentiva | Furosil; (RU) Russian Federation: Apo Furosemid | Furon | Furosemid | Furosemid milve | Lasix; (SA) Saudi Arabia: Apo furosemide | Diusemide | Drolina | Fusix | Lasix | Salurin; (SE) Sweden: Diural | Furix | Furoscand | Furosemid copyfarm | Furosemid hexal | Furosemid medical valley | Furosemid nordic drugs | Furosemid nycomed | Furosemid orifarm | Furosemid recip | Impugan | Lasix | Mindum; (SG) Singapore: Apo furosemide | Dirine | Dryptal | Frumid | Frusemide | Frusix | Furmide | Impugan | Lasix | Rasitol; (SI) Slovenia: Edemid | Furosemida gesfur | Furosemide s.a.l.f. | Fursemid | Lasix; (SK) Slovakia: Furorese | Furosemid | Furosemid bbp | Furosemid medreg | Furosemid slovakofarma | Furosemid xantis | Fursemid; (SR) Suriname: Apo furosemide | Frusemide | Furo denk | Fusid | Novo semide | Salurex; (SV) El Salvador: Furosemida PL | Furoset; (TH) Thailand: Aldic | Dema | Difu | Dimasix | Dirine | Diurine | Femide | Fluidsemide | Frusemide | Fruseno | Frusid | Fudirine | Furantral | Furetic | Furide | Furine | Furomide | Furosic | Furoside | Furotic | Fuseride | Fusesian | Hawkmid | Hydril-f | Impugan | Lasemide | Lasix | Lasix high dose | Laton | Losemide | Mediuresix | Mosix | Servisemide; (TN) Tunisia: Ansemid | Furolix special | Lasicare | Lasilix | Lasilix special; (TR) Turkey: Desal | Fuomid | Furotab | Lasix | Lizik | Tts furosemid; (TW) Taiwan: Ancoli | Anfuramide | Calitol | Combellin | Diosix | Edemalin | Excresin | F.S | Frusemide | Fuchonropin | Fucochu | Fulsix | Fumide | Furide | Furose | Furosely | Furosin | Furosix | Fursede | Fursemide | Fusemide | Huromide | Kofuzon | Lasemide | Lasix | Laside | Lysix | Nadis | Paiderton | Rasitol | Rosis | Schentese | Seridon | Solmatin | Suopinchon | Tochun | Ureside | Uretropic | Urex | Urian | Uroxin | Yidoli | Yusu; (UA) Ukraine: Frusemid | Furanthril | Furon | Furosemid | Lasix; (UG) Uganda: Agomide | Asix | Frusemide; (UY) Uruguay: Bios semide | Diusemide | Fischermide | Furoaster | Furosemida Lab. Chile | Furosemide | Furosemide efa | Furosemide Phs | Lasix; (VE) Venezuela, Bolivarian Republic of: Biosemida | Fromil | Furolax | Furosemida | Inclens | Lasix | Lifurox | Resimida | Salca; (VN) Viet Nam: Agifuros | Furostyl | Uloviz; (ZA) South Africa: Austell Furosemide | Beurises | Defulide | Dino-Retic | Furobe | Lasix | Merck-furosemide | Micro-furosemide | Mylan furosemide | Puresis | Rolab-furosemide | Uremide; (ZM) Zambia: Amedosix | Diasix | Frusemide | Furo denk | Furosimide | Kamfru; (ZW) Zimbabwe: Aquasin | Diuren | Fruside

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